Omar Ahmed

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Education

University of Oxford, *Oxford, UK* **10/2025–10/2029**
DPhil Engineering Science

University of Oxford, *Oxford, UK* **10/2021–06/2025**
MEng Engineering Science (Integrated Master's & Bachelor's)

Research Interests

Music recommender systems; user-based system evaluation; offline metrics for recommender systems; recommender system user studies. Research presently on music recommender systems focused on music discovery, with results to be generalisable to other domains. Current work centred on how a unified, user-centric understanding of discovery can be used to develop shared design criteria and evaluation protocols for recommender systems. Future work involves building a recommender that adheres to these criteria. Previous university-based and independent projects are at the intersection of music and technology.

Publications

O Ahmed. *Discover Weakly No More: User-Centric Design Criteria for Discovery in Music Recommendation*, 20th ACM Conference on Recommender Systems (RecSys), **27 September – 3 October 2026**, Minneapolis, Minnesota, USA.

O Ahmed, D De Roure. *Reducing The Geometric Complexity of Rooms For Simpler Acoustic Simulation*, 4th IEEE International Conference on Immersive and 3D Audio, **19–22 November 2026**, Venice, Italy. *(accepted pre-camera-ready manuscript provided)*

Relevant Experience

University of Oxford, *Oxford, UK* + **University of Birmingham**, *Birmingham, UK* **01/2026–Present**
Optimising Cultural Experiences for Mental Health in Underrepresented Young People Online
Recommender Research Consultant
- Creating a recommender, within a co-design environment with users, for recommendation of historical stories via a mobile application as a method to alleviate mental health issues in underrepresented young people.
- Collaborated with software developers in industry to inform recommender design.
- Developed experience in creating a recommender for public use, outside of only academic settings.
- Interviewed young people (users) within workshops to establish user requirements.
- Liaised with experts from a variety of backgrounds including psychiatry and engineering.
- Addressed practical concerns such as data and storage requirements of the application.

Queen Mary University London, *London, UK* **07/2024–01/2025**
Machine Listening Lab, Centre for Digital Music
Research Assistant
Title: *Improving Automatic Guitar Tablature Transcription with LLMs*
- Presented an *extended abstract* and *poster* at the 2024 Digital Music Research Network.
- Gathered data from existing datasets and neural network outputs to fine-tune a GPT-4o transformer.
- Discovered a key deficiency in current approaches and formulated a successful methodology that involved using LLMs to correct neural network outputs.

University of Oxford, *Oxford, UK* **10/2026–Present**
Designing Intelligent Musical Technology, Department of Engineering Science
Teaching Assistant
- Provide support to 3rd year undergraduate 'Designing Intelligent Music Technology' students.
- Assist in idea formation during early research, requiring an up-to-date understanding of the recent literature in the field.

- Cooperate with professors to prepare class plans and assess student work.
- Constructively critique research and offer insights into design solutions.
- Regularly support the students with any queries they have outside of class hours.

Relevant Academic Projects

Master's Thesis **10/2024–06/2025**

Title: *Reducing The Geometric Complexity of Rooms For Simpler Acoustic Simulation*

- Accepted manuscript for publication in the International Conference for Immersive and 3D Audio 2026.
- Developed an autoencoder-based model capable of identifying and maintaining the geometric complexity of perceptually relevant features of a room whilst smoothing less pertinent features.
- Combined knowledge of machine learning and musical room acoustics to create a novel framework for point cloud reconstruction.
- Contributed to the University of Oxford Bate Collection by simulating musical samples in historical spaces.

Bachelor's Thesis **10/2023–06/2024**

Title: *Automated application of a guitar tone according to a user-provided example tone sample*

- Leveraged WaveNet to design a system capable of detecting the guitar tone of a user-provided audio sample and applying the tone to the user guitar signal. This allows users to replicate the guitar tone of a recording without knowledge of the effects used.
- Investigated the viability of MFCC coefficients in quantifying the perceptual difference between guitar tones.
- Awarded £35,000 in funding after capitalising on findings to found a startup, achieving 1st place in the Reuben Entrepreneurship Forum and 2nd place in the Oxford Student Entrepreneurship Program.

Relevant Personal Projects

Investigating the link between brain waves and synth parameters **09/2024–04/2025**

- Measured EEG signals present whilst synth patches of various parameters are played.
- Investigated the mapping of brain wave types to synth parameters as a method to control a synthesiser.
- Collaborated with Dr Marcus Pearce, leader of the Music Cognition Lab at Queen Mary University London.

Stanford CCRMA Audio Effects Workshop **07/2023–09/2023**

- Engaged in the theory behind common audio effects such as reverb, distortion, and compression.
- Applied my knowledge in signal analysis and Python to musical applications.
- Created a multi-band compressor plugin in C++ using the JUCE framework.

Research Funding Awards

DPhil Funding, Oxford Graduate Scholarship EPSRC Doctoral Landscape Award, £110,000+ **2025–2029**

Research Grant, University of Oxford Rokos Scholarship, £1,500 **2023**

Skills

Application Software: *Advanced* – MATLAB, SolidWorks, EAGLE, Siemens NX

Programming Languages: *Advanced* – C++, Python (TensorFlow, PyTorch), MATLAB; *Intermediate* – SuperCollider

Languages: English (Native Speaker), Arabic (conversational)